Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)



SOLUTIONS ARCHITECT

Platform Product Owner | Program & Delivery Leadership
Product Strategy | Agile Delivery | Airline IT | Data Platforms

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Immediately available | Open to relocation | sefikegriboz.net

PROFESSIONAL SUMMARY

Key results: **98%** fewer reported ATC confusion events with an AI product | **EUR 8M** cumulative savings | ten years of **product ownership** on two **Tier-0** platforms | monthly incidents down from **8 to 1** on **six mission-critical systems** (1,000 users, **EUR 1M** budget)

Technical Product Owner and Program/Delivery Manager with 20+ years in airline IT, the last ten as de facto product owner of two **Tier-0** operational data platforms (TKAODS, OFPDS), with delivery and program accountability for operational platforms across scheduling, slot management, flight planning and IOCC (**EUR 6.5M** total program budget).

Turns operational problems into product outcomes: roadmaps, OKRs and KPIs for **six mission-critical systems** and user stories validated through UAT, on a high-transaction-volume integration estate of **200+ integration points**, **~50M annual transactions** and **99.8% availability**. Customer-side product owner for NetLine, SchedConnect and flight planning platforms across three carriers: a five-vendor RFP (3 shortlisted, 2 POCs) with business cases, and key-user training.

Product results in production: a fully AI-based callsign safety product for an **NP-hard** Quadratic Assignment Problem (**98%** fewer reported ATC confusion events), a real-time Early Warning System with zero restricted-airspace dispatch incidents, and **EUR 8M** cumulative savings. PSPO I and PMP certified, with **50+ projects** managed, the largest a **EUR 3.5M** program.

WORK EXPERIENCE

Career Break and Professional Development | Istanbul | Sep 2024 – Present

- Planned break after leaving the Turkish Airlines Group: family time, structured upskilling (**Harvard's CS50 AI**, 12 Python projects; **AWS Certified AI Practitioner**, **PSPO I** and **AWS Certified Cloud Practitioner** in 2026) and continued research on graph-based AI for aviation safety.

Principal Solutions Architect | Turkish Technology | Istanbul, Turkiye | Jan 2023 – Sep 2024

Official title: Distinguished Expert, the highest technical grade at Turkish Technology.

- Cut reported ATC confusion events by **98%**, from **50 to 1 per season** (sustained since), on up to ~1,500 daily peak-season flights, replacing the 2010 heuristic season-start assignment with an in-house fully AI-based callsign safety product aligned to EUROCONTROL CSST.
- Reframed callsign assignment as an **NP-hard** Quadratic Assignment Problem and built a scalable machine learning product on **Graph Neural Networks (GNN)**: **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering over phonetic, visual and semantic similarity.
- Reduced monthly incidents from **8 to 1** in 10 months as owner of product roadmaps, capability priorities, OKRs and KPIs for **six mission-critical operational systems** (1,000 users, ~100 concurrent, 5M annual transactions, EUR 1M budget).
- Balanced roadmap priorities against **integration dependency management** and architectural constraints.
- Ran the NetLine/Ops replacement selection (**5 vendors, 3 shortlisted, 2 POCs**; a **EUR 1.5M** decision) and vendor evaluation of flight planning platforms (Jeppesen, CAE formerly FPM, Lido) through RFP/RFI, vendor management and build-versus-buy business cases.
- Advised VP-level leadership on technology strategy and long-term product and R&D strategy.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers) that delivered **9 systems to production**, as architecture and design decision owner, with Scrum Master support as required.
- Defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers.
- Held schedule mismatches at **zero** across two carriers and two reservations systems during AJet's migration from TROYA, the group's reservations and inventory platform, to a separate third-party PSS, running SchedConnect-based synchronization, inter-carrier schedule distribution and schedule change management in a transition architecture.
- Shaped integration and technology strategy for next-generation aviation operations platforms as **NetLine Subject Matter Expert** for Turkish Airlines, SunExpress and AJet, and as SME in callsign safety, IOCC and operational control systems.
- Established design authority and design reviews bringing architecture decisions under ISO 27001 compliance, software quality management and enterprise architecture governance, and supported regulatory audits and ICAO compliance.

Senior Integration Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

Official title: Senior Expert, Integration Architect.

- Achieved **99.8% availability** (planned maintenance included) and **zero unplanned downtime** on mission-critical platforms serving **500+ concurrent users**, leading cross-functional delivery management across internal and external integration platforms through production deployment under high availability, disaster recovery design and SLA/SLO management.
- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model, with enterprise messaging and data streaming reference patterns adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Shortened new-integration lead time from **one week to one day**, cut integration-related incidents by **99%** and retired **47 legacy point-to-point interfaces** by reusing the group-wide reference patterns.
- Ran **Scrum**-based agile delivery with backlog management and prioritization in **Jira** and product documentation in **Confluence**: sprint planning and Definition of Ready/Done criteria that turned changing operational requirements into sprint-ready work.
- Enabled **1.5M** disrupted passengers a year to re-accommodate online, even those en route to the aircraft, by extending PSS/DCS integration with the in-house PSS TROYA to irregular operations (IROPS/IRROPS) and disruption management, propagating IOCC decisions to inventory, reservations and departure control as they happened.
- Eliminated dispatches into restricted airspace (**18** in the 12 months before go-live, **zero** since) by engineering a real-time Early Warning System across NetLine and FPM: **72** automated checks and alarms on planned routes for up to **~1,500 daily** peak-season dispatch legs.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions**.
- Applied microservices, enterprise integration patterns, API-first design and API lifecycle management, including naming conventions, API versioning and backward compatibility.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Raised release frequency from **1 to 4 a month** and **halved failed releases** by setting the estate's engineering and release practice: CI/CD architecture and DevOps release automation with Git-based (GitOps) configuration control, automated testing strategies, and performance and security testing gating releases.
- Caught **99%** of integration failures before users reported them and reduced mean time to detect from **one hour to one minute** with a real-time observability tool built across all live enterprise integrations.
- Based the tool on structured logging, correlation IDs and distributed tracing, and directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

Project Manager, Flight Operations | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

Functional title: Lead Solutions Architect and Platform Owner.

- Built **TKAODS** (Airline Operational DataStore) and **OFPDS** (Operational Flight Plan DataStore) into **Tier-0** 24/7 data platforms used by **950 users** across overseas offices, General Management and IOCC, in front of the Lufthansa Systems NetLine suite and flight planning system.
- Took schedule and flight-change delivery from **next-day to near-real-time** for **78 consuming systems**, replacing **70 separate flight-data interfaces** by routing every change through both platforms, 24/7.
- Fed operational changes (delays, cancellations) to passenger-notification systems, triggering **75M notifications a year**, and flight data to loyalty and disruption hotel systems, so all three ran on the same data as operations.
- Acted as product owner of TKAODS and OFPDS for **ten years** (2014 – Sep 2024), with final authority over feature prioritization and architectural evolution, translating operational requirements into user stories with acceptance criteria.
- Owned delivery scope, implementation planning, delivery dependencies, vendor activities, user acceptance testing (UAT), risk tracking and operational handover for multi-system releases across scheduling, flight planning, IOCC and downstream enterprise systems, through to production.
- Coordinated operational stakeholders and cross-team dependencies for two Tier-0 platforms, translating changing requirements into platform direction and carrying delivery through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.
- Owned enterprise-wide flight data distribution and a reporting layer on both platforms, serving **eleven main downstream domains** from one authoritative **single source of truth**: scheduling, crew, load control, PSS/DCS interfaces, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** (peak-season average) for up to **~1,500 flights a day**, 24/7 and near-real-time, across summer and winter seasons.
- Consolidated, transformed and enriched every update through source-to-target mappings, ETL/ELT pipelines and integration rules under data quality and metadata controls.

- Designed batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** flight-data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics.
- Set consumer-specific horizons, from a **rolling three-day window** to a **90-day window**.
- Designed the enterprise operational data and integration architecture on the AODB (Airline Operational DataStore) pattern, standardizing flight, leg, schedule and operational flight-plan data airline-wide: logical data models, entities and relationships, canonical data structures, and the ESB pattern applied twice.
- Managed NetLine's master and reference data (NetLine/Base) with the team: built and maintained the tables where users edit it and trained key users in master data management (MDM), keeping master data owned at its source.
- Drove technical direction, integration landscape and platform modernization for airline operational systems with a **EUR 6.5M** total program budget across scheduling, slot management, flight planning and IOCC, with delivery accountability as lead architect.
- Kept **850 NetLine users** working through the pandemic by coordinating, as NetLine product owner, the infrastructure teams' VMware solution.
- Received Turkish Airlines **achievement certificates** in 2017 for contributions to the Fuel Management Information System, the FPM & FE implementation and the OpsEye Diversion Manager module and its rollout.

Integration Solution Design Lead | **Turkish Airlines** | **Istanbul, Turkiye** | **Jan 2007 – Jun 2014**

Official title: Acting Team Leader.

- Delivered **EUR 8M** in cumulative savings from 2012 to 2023 (**EUR 800K** in the first year) with a multi-currency overflight cost engine built on a data pipeline of country entry/exit timestamps and segment distances.
- Led a **6-person team** through the lifecycle of **100+ integrations**, from requirements and system analysis to design, development, testing and documentation, harmonizing project plans and tracking cross-functional milestones per project.
- Earned PMP and ran scope definition, executive (C-level) stakeholder management, customer relationship management, vendor management, risk tracking, implementation planning and operational handover.
- Headed solution design and delivery for flight operations, crew and flight-planning integrations across global aviation vendors, building multithreaded real-time integration frameworks (2010) and contributing to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Introduced SchedConnect in 2007 for inter-carrier schedule enrichment and distribution, establishing recurring schedule data exchange with **54 partner carriers** under IATA data standards.
- Architected the schedule-to-PSS (TROYA) message integration under IATA SSIM Chapter 4 (SSM) and Chapter 5 (ASM) over Type B messaging, keeping inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated flights and disruption reason codes.
- Built **OFPDS** in 2010 to cut mainframe dependency: flight plans from the new planning system landed in Oracle and the data warehouse, easing SAP integration; the fuel efficiency team rated them far more accurate than the earlier Jeppesen feed.
- Developed the first heuristic Callsign Similarity & Assignment System (phonetic, visual and semantic similarity), delivered in 2010: the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by Turkish Airlines' **Chief Pilot and Head of Flight Operations** for flight operations projects, the callsign project first among them, and personally by the **General Manager and CEO**.

Software and Integration Engineer | **Turkish Airlines** | **Istanbul, Turkiye** | **Apr 2000 – Jan 2007**

Official title: Analyst Programmer.

- Contributed to the first production implementation of the Lufthansa Systems NetLine suite (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base), the platform that scaled with fleet growth from **60 to 500+ aircraft** and passengers from 10M (2000) to 85M (2024).
- Cut costs by **EUR 100K** a year by designing and building the timetable publication product from scratch: worldwide airline schedules and partner codeshare flights in, a PDF timetable for web download and print out, so the scheduling department began preparing the timetable books itself (Java, C, PL/SQL).
- Drove the technical work of the first major legacy modernization from day one as an implementation team member, moving schedule and operations control off the IBM mainframe (ADABAS/Natural) onto Oracle on Unix (AIX) and NetLine.

CORE COMPETENCIES & TECHNICAL SKILLS (hands-on)

Product Vision, Strategy and Discovery: Product vision and strategy, capability roadmaps, priorities, OKRs and measurable outcomes, product lifecycle management, product portfolio management, capability planning, requirements analysis, operational problem framing and requirements discovery with operations stakeholders, validated through UAT.

Program and Delivery Governance: Program management, delivery and transformation governance, implementation planning and milestones, dependencies, risk and issue tracking, vendor management, UAT, release management and operational readiness, budget oversight, cross-functional program coordination, executive (C-level) stakeholder management, PMI/PMP methodologies, workshop facilitation.

Outcome-Based Product Management and Agile Delivery: Capabilities prioritized against operational KPIs, reliability, safety and financial outcomes, with post-release impact measured; backlog refinement, Agile/Scrum, continuous improvement.

Airline IT Products: Integrated Operations Control Center (IOCC), operational and tactical flight planning (Jeppesen flight planning; FPM, now CAE Flight Planning Solutions; Lido), schedule management; DWH, SAP and MRP interfaces.

Technical Platforms, Data & Architecture: Enterprise Data Architecture, Data Modeling, Canonical Data Model, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture, Event-Driven Architecture, API Architecture, Apache Kafka, IBM MQ, Enterprise Service Bus (ESB), REST and SOAP APIs.

AI and Data Product Innovation: Neo4j and graph databases, PyTorch and PyTorch Geometric, MLOps, feature engineering, statistical analysis, data quality and metadata management.

Engineering Platforms & Languages: Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML; multithreaded real-time and database-driven systems; incident analysis and performance tuning.

EDUCATION

• Ph.D. (coursework)	Business Administration (Finance)	Yeditepe University	Istanbul
• M.Sc.	Management Engineering; thesis on data warehousing for mass customization	Istanbul Technical University	Istanbul
• B.Sc.	Computer Science Engineering	Istanbul University	Istanbul

LANGUAGES

• English	C1	Full Professional
• German	A2	Studying towards B1
• Turkish		Native

CERTIFICATIONS

• AWS Certified Cloud Practitioner (CLF-C02)	Amazon Web Services (AWS)	2026
• AWS Certified AI Practitioner (AIF-C01)	Amazon Web Services (AWS)	2026
• CS50's Introduction to Artificial Intelligence with Python (12 projects)	Harvard University (CS50)	2026
• Professional Scrum Product Owner I (PSPO I), Credential ID 1347372	Scrum.org	2026
• AI and Cloud Foundations: OCI AI Foundations; Agentic AI Foundations; IBM SkillsBuild; PMI AI (CPMAI, GenAI)	Oracle, IBM, PMI	2025
• Docker Foundations Professional Certificate	Docker	2025
• Manager Ready (DDI) leadership simulation: exceptional rating	TAG Taurus Group	2024
• ITIL v3 Foundation, Candidate No. 990453483	BCS, The Chartered Institute for IT	
• Project Management Professional (PMP), Credential ID 1225730	PMI	
• Lufthansa Systems SchedConnect vendor training (two weeks)	Lufthansa Systems	